
Pop Stars Have Something to Teach CU Professors: How Low Can You Go with Sentence Embeddings to Stay Useful?

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Abstract

How much embedding dimensionality is needed for a given corpus? Can a model trained on a simpler domain transfer effectively to semantically richer tasks? How does sensitivity to embeddings dimensionality vary across corpora?

To answer these questions, we study how domain and embedding dimensionality interact in unsupervised sentence encoders. Using two extreme corpora, **Taylor Swift lyrics** and **Nakul Verma’s papers**, we train **contrastive BERT models from 4D to 767D** and evaluate geometry and transfer learning properties.

Surprisingly, lower-dimensional **models trained on lyrics (4-256D) outperform paper-trained models on STS-B**, with papers only catching up at 767D. This suggests that simple, high-frequency text can be an efficient pretraining source for compact encoders, while complex text needs far more dimensions to expose useful structure.

1 Introduction

Sentence embeddings are mappings from high-dimensional linguistic signals to lower-dimensional structures whose geometry captures the semantic meanings of sentences. Latent space embeddings are crucial in NLP tasks. Recent progress—including BERT Devlin et al. [2019] for token embeddings and then Sentence-BERT Reimers and Gurevych [2019] for sentence embeddings—has significantly improved sentence embedding quality. In practice, resulting embeddings are typically used per se (in 768D or 1024D for BERT-based models) without any consideration given to decreasing the compute or memory on downstream tasks via a lower embedding dimensionality. However, the nature of the data itself and the downstream task might require a different dimensionality, which could make the resulting embeddings not only more efficient but also useful.

We train sentence representation models using a two-step procedure and evaluate their ability to generalize across domains. Our evaluation steps include 1) assessing transfer learning of the models trained on one particular dataset (either pop songs or scientific papers) on general, out-of-distribution tasks, and 2) the resulting geometry of in-domain and out-of-domain data embedded with our trained models. Through this analysis, we can also infer the intrinsic dimensionality of the topic-specific data used to train the models under evaluation.

On a high level, we take a step towards answering the following questions: Which conceptual domains (represented by the data used to train our models) allow for smaller-dimensional representations that still capture meaning and retain transfer learning abilities? How much can we decrease the embedding dimension while retaining useful representations?

Related Work

Widespread adoption of transformer encoders began with BERT Devlin et al. [2019], which learns deep contextual token representations that can be repurposed for sentence-level meaning. Sentence-BERT Reimers and Gurevych [2019] extends BERT with a siamese architecture and pooling scheme, enabling efficient cosine-similarity retrieval and establishing a strong supervised benchmark for sentence embeddings. Li et al. [2020] analyze sentence embeddings produced by pretrained language models, showing strong anisotropy and proposing geometric corrections to improve isotropy and downstream task performance.

Contrastive learning provides the main algorithmic backbone for modern sentence encoders. Wang and Isola Wang and Isola [2020] show that contrastive objectives implicitly trade off two geometric pressures—alignment of positive pairs and uniformity of embeddings on the hypersphere—offering a framework we directly use to interpret our models. Wang and Liu Wang and Liu [2021] further study the behavior of contrastive losses such as InfoNCE under different design choices, explaining how temperature, sampling, and margin parameters induce distinct embedding geometries. Building on these ideas, SimCSE Gao et al. [2021] introduces a simple but powerful contrastive objective for sentences, where dropout-generated views serve as positive pairs and hard negatives emerge naturally from in-batch sampling.

Recent work also focuses on the dimensional structure of sentence representations. Wang et al. Wang et al. [2023] study the performance of low-dimensional sentence embeddings from pretrained encoders, decomposing the impact of the encoder and pooler and showing that substantial dimensionality reduction is often possible without dramatic loss in quality. Amit et al. Amit et al. [2021] investigate how intrinsic dimensionality governs generalization, arguing that effective models adapt their representation dimensionality to the complexity of the underlying data manifold—a perspective that motivates our focus on how different corpora support useful low-dimensional sentence embeddings.

2 Methodology

We use contrastive learning described in SimCSE Gao et al. [2021]. SimCSE’s training objective is to pull positive pairs together in embedding space, and push negative pairs apart. In our context, positive and negative pairs are defined by closeness in the same text, creating self-supervised setting. SimCSE was previously shown successful on STS benchmarks and transfer tasks, which is why we use it as a main semantic meaning learning method.

We training two architectures: encoder + pooler (BERT + feed-forward NN), encoder + projection (BERT + PCA).

2.1 Encoder + pooler

Here we briefly discuss implementation details, and all code is available.

The encoder maps tokens to latent representations, and we use BERT for that. The pooler maps hidden states to low-dimensional embedding of sentences.

In Wang et al. [2023] found that performance loss can be decomposed into separate encoder and pooler contributions, which is why we use two-step training procedure. First, we freeze parameters of pooler and fine-tune the encoder, achieving best version of BERT for this dataset; then, we freeze the encoder and train only pooler. This allows to reduce encoder loss and pooler loss sequentially.

We load ‘bert-base-uncased’ weights from Hugging Face for encoder Devlin et al. [2019], and construct a fully connected network with hidden-size x hidden-size layer, tanh activation, hidden-size x projection-size fully-connected layer (following Wang et al. [2023]). We apply mean pooling to aggregate token embeddings into sentences embeddings; note that BERT produces position-aware token representations, so ordinal information is not lost.

We train embeddings as follows. We apply SimCSE trick: a sentence passed as it and with dropout

should have similar embeddings. For each anchor sentence, we find multipositive Normalized Temperature-scaled Cross-Entropy loss and optimize using Adam.

2.2 Encoder + projection

Encoder is BERT, followed by PCA. For given projection dimensionality k , we take first k principal components.

2.3 Base model

BERT-base-uncased with mean pooling. A projection head maps pooled representations to a target dimension $d \in \{4, 16, 64, 256, 767\}$ followed by L_2 normalization.

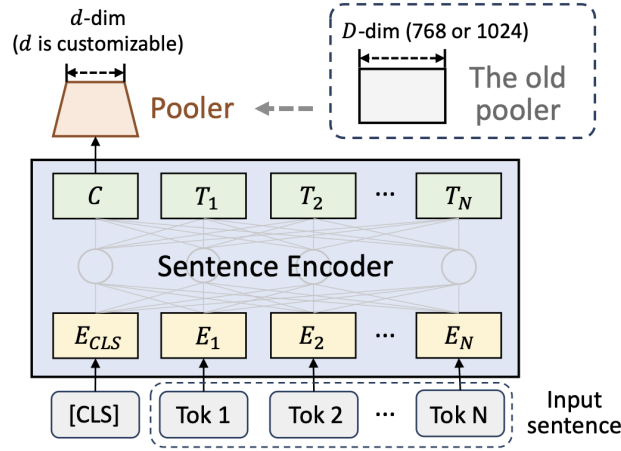


Figure 1: BERT+Pooler architecture. Image credit: <https://arxiv.org/pdf/2310.15285>.

2.4 Training domains

We choose domains to capture different styles, word frequency, mode of communication. Given these are opposite fields, ability to capture both indicates strong ability to generalize across topics in language.

Each song and each paper section is treated as one document, and neighboring sentences within a document form positive pairs. The corpora are token-matched for training compute parity:

- 1) Taylor Swift lyrics (short, repetitive, low-entropy)
- 2) Nakul Verma’s ML papers (long, information-dense)

We split text into sentences, which we consider smallest meaningfully-complete unit. Pre-processing methodology is discussed below.

2.5 Data preprocessing

Downloading, cleaning, and formatting the datasets into sentence-level documents was a multistep process.

For the lyrics dataset, there was a kaggle repository containing the entire Taylor Swift discography, which we used for our lyrics corpus. The cleanup included removing sequential duplicate sentences, filtering out song and album metadata, and splitting the lyrics into individual sentences using the NLTK sentence tokenizer. Each newline was treated as its own sentence.

For the papers dataset, we used the arXiv API to download Nakul Verma’s papers in pure latex form. There was substantial cleaning and processing work involved from that point. First, we removed latex commands, equations, figures, tables, and references using regex patterns. Next, we split the cleaned

text into sentences. These two actions required a fair amount of user judgement, and diminished the size of the usable corpus.

Given two corpora, we create pairs of similar sentences in self-supervised manner through assigning similarity indicator based on sentence distance.

2.6 Training objective

We train unsupervised SimCSE-style contrastive models with multiple positive examples, sampling local neighbors within each document. Following Wang et al., 2023 Wang et al. [2023], we use a two-step training scheme:

Step 1. Freeze pooler, train encoder (3 epochs).

Step 2. Freeze encoder, train pooler (3 epochs).

2.7 Evaluations

- Geometry
 - Alignment and uniformity losses (Wang et al., 2020, Wang and Isola [2020]) on in-domain and out-of-domain test sets. Alignment measures how close positive pairs are, and uniformity captures how evenly embeddings are spread on the hypersphere.
 - t-SNE embeddings of sentences from songs and papers.
- Transfer learning: STS-B Spearman rank correlations of sentence pair embeddings.

PCA projections of the frozen BERT encoder at the same target dimensions are used as a baseline.

3 Results and Analysis

3.1 Spearman Rank on STS-B

The results on lyrics-trained models rise sharply between 4D and 256D, peaking long before the full BERT size of 768D (see Fig. 2). Paper-trained models, in contrast, only come to full potential at 767D, hinting at a much **higher intrinsic dimensionality of scientific text**. Since STS-B is out-of-distribution for both training corpora, the gap between the performance of songs- and papers-trained models cannot be attributed to lyrics being much "closer" to STS-B than papers. Instead, it reflects how contrastive fine-tuning under a low-dimensional bottleneck **reallocates capacity of the model**.

On the lyrics corpus, the contrastive task is relatively simple and repetitive, so the model can solve it without drastically reshaping the pre-trained BERT geometry. A small projection head is enough to compress lyrics-specific structure while preserving generic semantic directions that still transfer to STS-B.

On the papers corpus, the contrastive task is much harder: the model must separate dense, technical sentences within and across sections. In low dimensions, the encoder and head are forced to specialize for fine-grained distinctions, which distorts general-purpose semantics and hurts STS-B transfer. Only when we increase the projection size up to 256-767D does the paper-trained model regain enough capacity to encode both in-domain nuance and broad semantic structure, at which point it overtakes the lyrics-trained model.

The PCA baseline stays near zero Spearman rank across all projection sizes, suggesting that **linear compression alone cannot expose semantic structure** – the pooler must learn it.

3.2 Uniformity and Alignment

All plots 3 show the **alignment vs uniformity tradeoff**: higher dimensions improve uniformity but degrade alignment. For PCA specifically, uniformity losses are much smaller than alignment losses for all dimensions, indicating that embedded sentences occupy the full hypersphere but do not preserve local similarity structure.

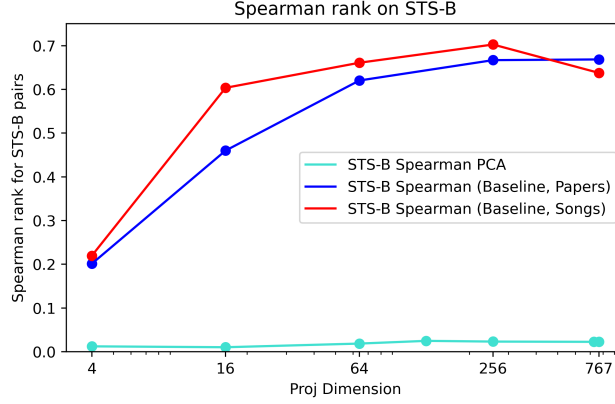


Figure 2: Spearman rank correlations for BERT+pooler and BERT+PCA variants.

For contrastive training, **in-domain data** (eg. songs embedded with a model trained on songs) **becomes well-spread** (good uniformity) but not overly tight, and **out-of-domain data** often collapses, achieving **artificially low alignment with poor global geometry**. This collapse likely occurs because the model is forced to project unfamiliar sentence structure into a space shaped only by in-domain positives, over-compressing local neighborhoods without preserving meaningful global structure. In addition, papers-trained models behave more similarly in terms of both losses on different domains, while songs-trained models have larger oscillations when domains are switched, hinting at more stable geometry learned by papers-based embeddings across domains.

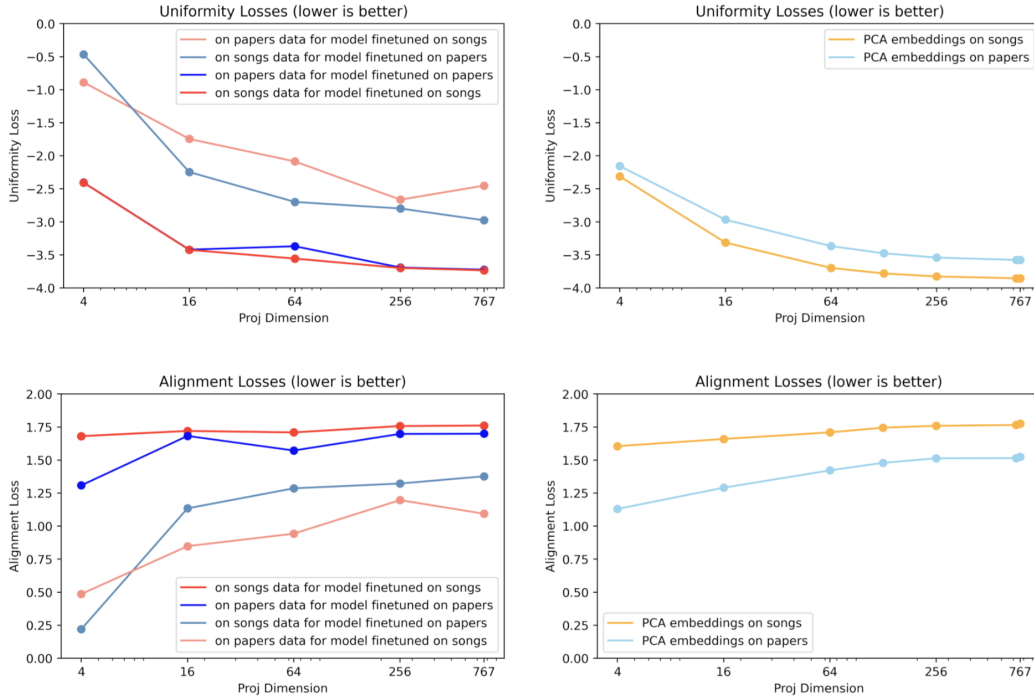


Figure 3: Uniformity and alignment losses.

3.3 Songs and Papers Clusters Embedded with t-SNE

Even with only 4 dimensions, the embeddings already separate lyrics from papers as two distinct clusters (see Fig. 4), indicating that coarse domain identity survives aggressive compression. However,

in 4D the clusters are elongated and less compact, with visibly distorted shapes. At 16 dimensions, the clusters become rounder, and local neighborhoods look more stable. This suggests that increasing dimensionality from 4D to 16D does not create domain separation from scratch, but instead improves the within-domain geometry: it reduces compression artifacts and yields a smoother manifold, which is consistent with the concurrent improvement in STS-B performance.

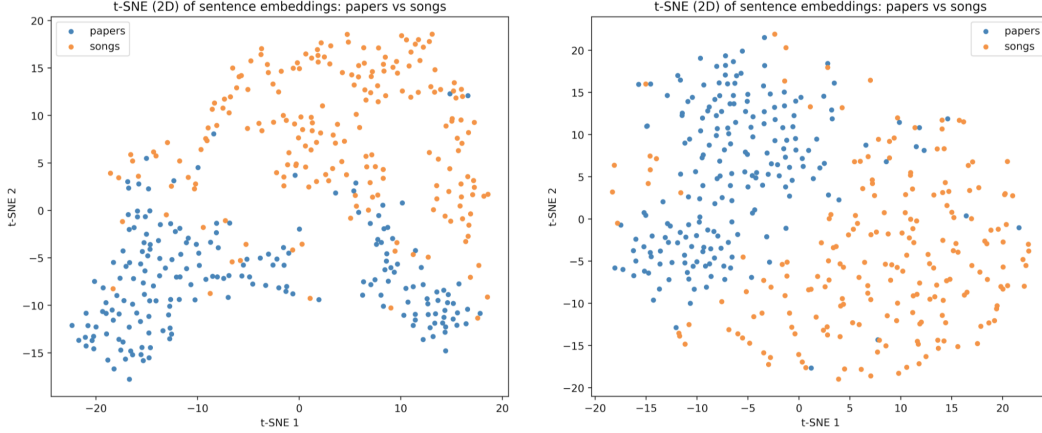


Figure 4: t-SNE results for 4D and 16D BERT+pooler models trained on songs.

3.4 Ablation on the Training Procedure

For songs-trained models, with smaller dimensions (4, 16), the two-step training process yields the best Spearman rank correlations, showing how **important the pooler is for extreme dimensionality reduction** (see Fig. 5). However, at certain higher dimensions training only the encoder yields the best results. Training only the pooler is not sufficient for learning useful representations for songs-based models.

For papers-based models, we see a similar trend, but the dimension cutoff at which the pooler stops giving much benefit is higher, somewhere between 128D and 256D (again hinting at the higher intrinsic dimensionality of scientific data). In addition, only training the pooler with the frozen BERT backbone yields much better Spearman rank results for papers-trained models than for songs-trained models – perhaps hinting to us that the frozen BERT encoder is initially better suited for academic text.

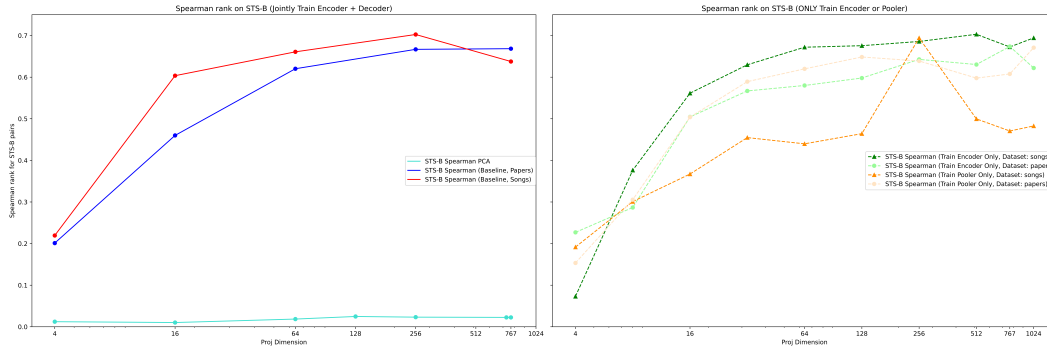


Figure 5: Ablation study on the two-step training procedure.

3.5 Challenges and Limitations

We note that all reported results depend on a specific choice of backbone (BERT-base), corpus size, and contrastive neighborhood definition. In addition, while STS-B Spearman rank results provide an insightful semantic probe, it represents only one notion of transfer learning performance.

Evaluation Criteria Selection of evaluation metrics (alignment, uniformity, STS-B Spearman rank) was an area of much thought and discussion. Drawing inspiration from Gao et al. [2021], we explored enriching our unlabeled dataset with self-supervised positive pairs. We landed on sentence proximity as a straightforward and effective method to generate these pairs without additional labeling.

Contrasting Loss Finetuning By design, the contrastive loss finetuning only seeks to optimize local structure and better distribute embeddings along the unit hypersphere. However, the inherent upper bound on this optimization objective comes from the pre-trained representations provided by BERT. Our procedure is not designed to, nor is it likely to, drastically alter the global structure of the embedding space. Hence, this route will not likely yield a dramatic improvement in sentence embedding, or revolutionize the field.

Scope Due to time and resource constraints, we limited our experiments to two esoteric domains of the English language. A more comprehensive study would include a much broader array of domains, self-labeling algorithms, and backbone architectures.

BERT As our backbone, BERT-base-uncased is a well-established model for sentence embeddings. However, there are more recent encoder architectures (e.g., RoBERTa, DeBERTa, etc.) available for exploration. In addition, the past five years have seen a preponderance of investment into closed-source LLMs (e.g., GPT, Gemini, Grok, Claude, etc.) that may offer superior embedding capabilities (with the caveat that it's uncertain that cleanly demarcated encoder-decoder patterns exist in said closed models).

3.6 Future Directions

For future work, we would like to explore **additional domains** (such as news, code comments etc.), try other **embedding models beyond BERT**, use **alternative training objectives** (for instance, non-contrastive methods), enrich the suite of **evaluation methods**, and implement competing self-supervised **sentence pair labeling algorithms**.

Expanding the scope a bit, it would be interesting to move beyond all-purpose sentence embeddings and explore **task-specific embeddings** and **multimodal embeddings**, modeling language in a **hierarchical manner**, moving beyond Euclidean spaces into **non-Euclidean geometries**, and exploring the notion of **intrinsic dimensionality**.

4 Conclusion

4.1 Summary of Findings

- **Lyrics-trained models dominate at low dimensions (4–256D) on STS-B**, while papers-trained models only overtake at 768D $\Rightarrow$ **scientific text appears inherently higher-dimensional**.
- **Uniformity and alignment conflict** with each other, with **papers-trained models being more stable** when transitioning from in-distribution to out-of-distribution data.
- **Low-dimensional embeddings** are able to **separate domains (even at 4D)**, but more meaningful local structure emerges only at 16D or more.
- Linear compression (**PCA**) is **insufficient for transfer learning** tasks in our experiments: contrastive learning is necessary to expose semantic structure even at small dimensions. It is advised to retrain, instead of linearly transform dimensions, if target dimensionality changes.

4.2 Contributions of each team member

- **A.C.:** Partial preprocessing, BERT+pooler contrast-and-freeze architecture implementation, BERT+PCA architecture implementation and evaluation on 4-767D, partial report write-up.
- **A.K.-W.:** Data collection and preprocessing for both datasets, ablation studies on the BERT+pooler architecture for both datasets with training and evaluation on 4-767D, partial report write-up.

- **M.G.:** Initial literature review, evaluation suite and graphics implementation, baseline BERT+pooler training and evaluation on 4-767D for both datasets, poster preparation, partial report write-up.

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Code Availability

All our code can be found at <https://github.com/AlenaResiko/UML>.

References

- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4171–4186. Association for Computational Linguistics, 2019. URL <https://arxiv.org/abs/1810.04805>.
- Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 3982–3992. Association for Computational Linguistics, 2019. URL <https://arxiv.org/abs/1908.10084>.
- Bohan Li, Hao Zhou, Junxian He, Mingxuan Wang, Yiming Yang, and Lei Li. On the sentence embeddings from pre-trained language models. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 9119–9130. Association for Computational Linguistics, 2020. URL <https://arxiv.org/abs/2010.08240>.
- Tongzhou Wang and Phillip Isola. Understanding contrastive representation learning through alignment and uniformity on the hypersphere. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119, pages 9929–9939. PMLR, 2020. URL <https://arxiv.org/abs/2005.10242>.
- Feng Wang and Huaping Liu. Understanding the behaviour of contrastive loss. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2495–2504, 2021. URL <https://arxiv.org/abs/2012.09740>.
- Tianyu Gao, Xingcheng Yao, and Danqi Chen. Simcse: Simple contrastive learning of sentence embeddings. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 6894–6910. Association for Computational Linguistics, 2021. URL <https://arxiv.org/abs/2104.08821>.
- Hongwei Wang, Hongming Zhang, and Dong Yu. On the dimensionality of sentence embeddings. *arXiv preprint arXiv:2310.15285*, 2023. URL <https://arxiv.org/abs/2310.15285>. Tencent AI Lab Seattle.
- Yali Amit, Michael Fink, Nathan Srebro, and Shimon Ullman. The role of intrinsic dimensionality in generalization. In *Advances in Neural Information Processing Systems*, volume 34, pages 23125–23137, 2021. URL <https://arxiv.org/abs/2104.05693>.

Appendix